MIL-T-27736A
28 February 1968

SUPERSEDING
MIL-T-27736(USAF)
6 May 1963

MILITARY SPECIFICATION

TAPE, TEXTILE, NYLON, FOR RING-SLOT PARACHUTES

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of nylon tape.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

CCC-T-191	Textile Test Methods

Military

MIL-W-43334	Webbing and Tape, Textile, Packaging and Packing of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8315

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3. REQUIREMENTS

3.1 Materials

3.1.1 Yarn. The yarn shall be of 840/140 denier, a bright, high tenacity, light and heat resistant polyamide prepared from hexamethylene diamine and adipic acid, or its derivatives. It shall have a minimum melting point of 472°F. The yarn shall not be bleached in any manner or in any subsequent process.

3.1.1.1 Twist. The warp and filling yarn shall have a minimum of 2-3/4 turns of "Z" twist per inch.

3.2 Weave

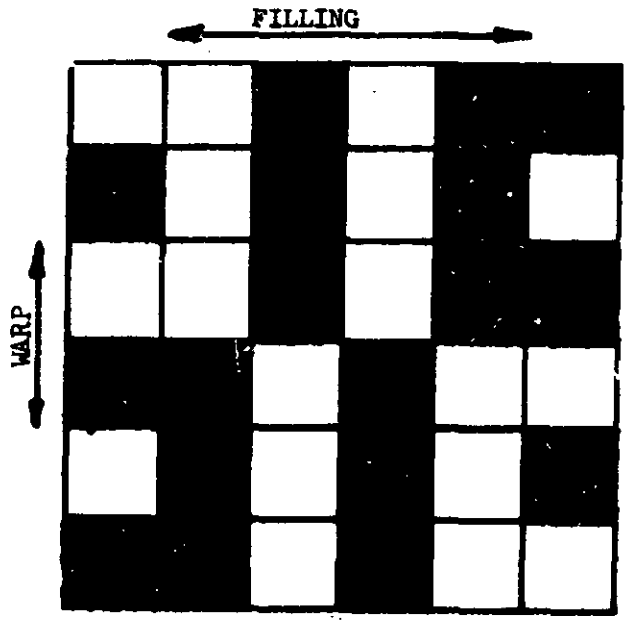
3.2.1 Pattern. The body and selvage weave shall be in accordance with figure 1. The selvage weave shall be 2 ends of 840/140 denier woven as 1.

3.3 Construction and physical properties. The finished tape shall conform to the requirements specified in table I and 3.3.1 through 3.3.3.

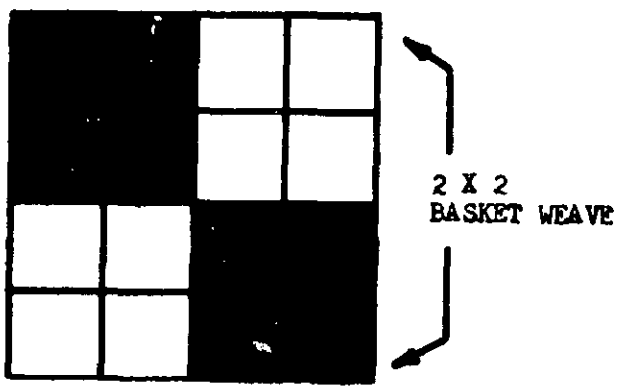
Table I. Construction and Physical Properties

Characteristic	Requirement
Weight, ounce/linear yard (maximum)	3.5
Yarns, number per inch (minimum)	
Warp	48
Filling	36
Selvage	41
Total tape width (including selvages), inches	11 ± 1/4
Selvage width (each selvage), inches	1-1/8 ± 1/16
Breaking strength, pounds/inch of body width (minimum)	
Warp	625
Filling	450
Breaking strength, warp, pounds/inch of selvage width (minimum)	900
Acidity-alkalinity (pH)	4.5 to 8.5
Permanence of finish	
Air permeability, cu ft/min./sq ft	70 ± 20
Thickness increase, percent (maximum)	10
Shrinkage, percent (maximum)	
Warp	5
Filling	5

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BODY



SELVAGE

FIGURE 1. Weave Patterns

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3.3.1 Color. The tape shall be in natural only.

3.3.2 Finish. The finished tape shall be smooth and even and shall not contain sizing, lubricating or weighting materials.

3.3.3 Resistance to light and heat. The tape shall lose no more than 25 percent of its original breaking strength when tested for resistance to light and to heat as specified in 4.3.2 and 4.3.3.

3.4 Length and put-up. The tape shall be put up in rolls. Unless otherwise specified, each roll shall contain not more than 3 pieces and the minimum length of any piece shall be 10 yards. The length of the roll shall be not more than 110 yards nor less than 90 yards.

3.5 Identification tickets. Each roll of tape shall have an identification ticket attached to the roll in accordance with MIL-W-43334. The number of pieces in the roll shall be marked on the ticket.

3.6 Fiber identification. Each roll of tape shall be labeled, ticketed, or invoiced for fiber content in accordance with Textile Fiber Products Identification Act.

3.7 Workmanship. The finished tape shall be clean, evenly woven, and shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Except when otherwise specified, inspection shall be in accordance with the provisions of MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise

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excluded, amended, modified or qualified in this specification or applicable purchase documents. In addition, components and materials listed in table II shall be tested for the characteristics specified and in accordance with the referenced test methods of CCC-T-191. The lot size shall be expressed in pounds, and the sample unit shall be 500 yards of the nylon yarn. The lot shall be unacceptable if one or more units fail to meet any requirement specified. All test reports shall contain the individual values utilized in expressing the final result. The sample size shall be as follows:

Lot size (pounds)	Sample size
800 or less	2
801 thru 2,000	3
2,001 and over	5

Table II. Testing of Components

Characteristic	Requirement paragraph	Test method (CCC-T-191)
Warp and filling yarn		
Nylon identification	3.1.1	1530 1/
Melting point	3.1.1	1534
Denier	3.1.1	4021 1/
Twist	3.1.1.1	4052 or 4054 1/

1/ The supplier's certificate of compliance shall be accepted for this characteristic. When required the certificate shall be accompanied by actual test, inspection or other verifiable data.

4.2.2 Inspection of the end item

4.2.2.1 Examination of the end item. Defects found during the examination shall be classified in accordance with 4.2.2.1.1 through 4.2.2.1.3.2.

4.2.2.1.1 Yard-by-yard examination. The required yardage of each piece shall be inspected on both sides and visual defects classified as listed in table III. All defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the tape, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The sample unit for this examination shall be one linear yard. The sample size shall be in accordance with level III of MIL-STD-105. The acceptable quality level shall be 1.0 major and 6.5 total (major and minor combined) defects per 100 units (yards). The lot size shall be expressed in units of one linear yard each. An approximate equal number of yards shall be examined from each roll selected. The number of rolls from which the sample is to be selected shall be in accordance with table IV of this specification.

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Table III. Classification of Defects

Characteristic	Defect	Major	Minor
Abrasion mark	Any abrasion mark showing fuzziness.	X	
Biased filling	More than 2 inches from horizontal at greatest point of bias.	X	
Broken or missing end	Two or more contiguous regardless of length.	X	
	Single, more than 18 inches missing.	X	
	Single, 18 inches or less missing.		X
Broken or missing pick	Two or more contiguous regardless of length.	X	
	Single, regardless of length.		X
Coarse filling bar	Clearly noticeable 1/ and extending for more than 1 inch in the length direction of the cloth.	X	
	Clearly noticeable 1/ and extending for 1 inch or less in the length direction of the cloth.		X
Crease	Hard, embedded crease	X	
Cut, hole, or tear	Three or more warp or filling threads ruptured at adjoining points.	X	
Distortion or slippage of threads	Any distortion or slippage of warp or filling threads that cannot readily be reset by hand.	X	
Filling bow	More than 2 inches in height (measured from a straight line cord to highest point of arc).	X	
Fine filling bar	Any clearly noticeable 1/ fine filling bar.	X	

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Table III. Classification of Defects (Cont'd)

Characteristic	Defect	Major	Minor
Floats or skips	Any multiple float more than 3/16 inch in either warp or filling direction. Single floats more than 1/4 inch in length.	X	
	Pin floats, sequence of more than 1 inch (continuous) in length.	X	
	Two or more multiple floats (in any linear yard) more than 1/8 inch in either warp or filling direction.		X
	Single floats more than 1/8 inch but not more than 1/4 inch in length.		X
	NOTE: Single floats 1/8 inch or less shall not be considered as defects.		
Heavy filling bar or heavy place	Clearly noticeable 1/ and extending for more than 1 inch in the length direction of the cloth.	X	
	Clearly noticeable 1/ and extending for 1 inch or less in the length direction of the cloth.		X
Hitchback (warp catch)	Resulting in a thin place 3/8 inch or more in combined warp and filling directions.	X	
Jerked in filling or slough-off	Two or more additional yarns in the shed.	X	
	One additional yarn in the shed.		X
	NOTE: One-half inch or less shall not be considered a defect.		
Loops, kinks, or snarls, (except selvage)	Any more than 1/8 inch in length.	X	
	Three or more (in any linear yard) 1/8 inch or less in length.	X	
	One or two (in any linear yard) 1/8 inch or less in length.		X
Mispick or double pick	Two or more additional picks in the shed.	X	
	One additional pick in the shed.		X

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Table III. Classification of Defects (Cont'd)

Characteristic	Defect	Major	Minor
Pick out mark	Resulting in a clearly noticeable thin or light place.	X	
Selvage cut, broken, torn, or scalloped	Any cut, broken, torn, or scalloped selvage.	X	
Selvage slack or wavy	Clearly noticeable 1/ waviness along selvage edge when viewed without tension.	X	
Selvage stringy or loopy	Any stringy or loopy selvage projecting more than 1/8 inch.	X	
	More than 3 inches of continuous stringy or loopy selvage projecting 1/8 inch or less.	X	
	Three inches or less of continuous stringy or loopy selvage projecting 1/8 inch or less.		X
Selvage tight	Any clearly noticeable 1/ roll of edge or edges when tension is released.	X	
Smash	Any smash.	X	
Spot, stain or streak	Clearly noticeable 1/and more than 1/2 square inch in area.	X	
	Clearly noticeable 1/but not more than 1/2 square inch in area.		X
Strip back	More than 5 per linear yard over 1/4 inch in length.	X	
	Up to 5 per linear yard over 1/4 inch in length.		X
Thin or light place, light set mark	Any clearly noticeable 1/ reduction in filling texture.	X	
Undrawn filling yarn (caused by uneven filling shrinkage)	Resulting in clearly noticeable 1/ pucker.	X	
Weak places	Any weak place.	X	

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Table III. Classification of Defects (Cont'd)

Characteristic	Defect	Major	Minor
Wrong draw	Resulting in a clearly noticeable 1/ warpwise streak more than 18 inches in length.	X	

1/ The term "clearly noticeable" as contained in the defect description shall be interpreted to mean clearly visible at normal inspection distance (approximately 3 feet).

Table IV. Sampling Plan

Lot size (yards)	Sample size (rolls)	Maximum number defects acceptable in sample (see 4.2.2.1.2 thru 4.2.2.1.3.2)
Up to 1200 1/	3	0
1201 up to and including 3200	5	0
3201 up to and including 10,000	8	0
10,001 up to and including 35,000	13	0
35,001 up to and including 150,000	20	1
150,001 and over	32	2

1/ If a lot contains fewer than 3 rolls, each roll in the lot shall be examined.

4.2.2.1.2 Overall examination. Each defect listed below shall be counted no more than once in each roll examined. The sample unit for this examination shall be one roll. The sample size and acceptance number shall be as shown in table IV.

Defects

Objectionable odor.
Number of pieces in roll not indicated on piece ticket.
Uncleanliness throughout.
Uneven weaving throughout.
Not labeled in accordance with Textile Fiber Products Identification Act.

4.2.2.1.3 Length examinations

4.2.2.1.3.1 Individual rolls. Each roll in the sample shall be examined for the defects listed below. The sample unit for this examination shall be one roll. The sample size and acceptance number shall be as shown in table IV.

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Defects

Gross length less than specified minimum length or more than specified maximum length.
Gross length more than 2 yards less than gross length marked on piece ticket.
Any roll containing more than 3 pieces.
Any piece less than 10 yards in length.

4.2.2.1.3.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual gross length of rolls in the sample is less than the total of the gross lengths marked on tickets.

4.2.3 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing and marking requirements of section 5 are in accordance with MIL-W-43334.

4.3 Testing of the end item. The methods of testing specified in CCC-T-191, wherever applicable,and as listed in table V shall be followed. Except as otherwise specified, all tests shall be performed under standard conditions in accordance with CCC-T-191. All tests reports shall contain the individual values utilized in expressing the final result. The sample size shall be as follows:

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

The lot shall be unacceptable if one or more units fail to meet any requirement specified. The lot size shall be expressed in units of 1 linear yard. The sample unit for testing shall be 15 yards of tape.

Table V. Test Methods

Characteristic	Requirement reference	Test Method (CCC-T-191)
Weight	Table I	5040
Yarns per inch		
Warp	Table I	5050
Filling	Table I	5050
Width		
Overall	Table I	5020
Selvage	Table I	5020

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Table V. Test Methods (Cont'd)

Characteristic	Requirement reference	Test Method (CCC-T-191)
Breaking strength		
Warp	Table I	5104
Filling	Table I	5104
Selvage	Table I	1/ 5104
Nonbleaching	3.1.1	2/
Weave	3.2	Visual 3/
Acidity-alkalinity (pH)	Table I	2811
Permanence of finish		
Air permeability	Table I	5450 and 4.3.1
Thickness	Table I	5030 and 4.3.1
Shrinkage		
Warp	Table I	4.3.1
Filling	Table I	4.3.1
Light resistance	3.3.3	4/ 4.3.2
Heat resistance	3.3.3	4/ 4.3.3

1/ Except the specimen width of 1-1/2 inches does not apply.

2/ This requirement shall be accepted on the basis of the supplier's certificate of compliance with specification requirements.

3/ One determination per sample unit and results reported as "pass" or "fail."

4/ Specimens shall be selected from the continuous warp yarns as the specimens for the original breaking strength tests.

4.3.1 Permanence of finish. The tape shall be conditioned for a period of 4 hours under standard atmospheric conditions specified in section 4 of CCC-T-191. A 9-inch square, excluding the selvages, shall be marked on the conditioned specimen by means of a template, using indelible ink. Determine the thickness of the tape, both body and selvage. A metal tub of adequate size to accommodate the specimen, prepared as described

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below, shall be filled to within 3 inches of the top with water which shall then be heated to a boil. The specimen shall be placed in the boiling water in a "skein" form prepared by stapling two opposite sides of the specimen together to form a loop or skein. This shall then be placed over a glass rod 1/4 inch in diameter and 21 inches in length. A glass tube 1/4 inch in diameter and 21 inches long weighing approximately 45 grams shall be placed inside the loop at the bottom. The loop shall then be suspended in the boiling water bath by attaching it to another glass rod 24 inches long and 1/4 inch in diameter by means of twine or wire. The 24-inch glass rod shall rest on the top of the tub allowing the specimen to hang freely in the bath. The specimen shall be subjected to the action of the boiling water bath for a period of 15 minutes after which it shall be removed from the tub and allowed to drain for a few minutes, and the staples shall be removed from the sample which shall then be placed flat on a horizontal screen to dry. After the specimen is thoroughly dry, it shall be exposed for at least 4 hours to a standard atmosphere of 65 ± 5 percent relative humidity and a temperature of 70° ± 2°F. The 9-inch square shall then be measured in three places in the warp direction and three places in the filling direction, and the amount of shrinkage recorded. Air permeability and thickness tests shall again be conducted on the specimen to determine the air permeability and increase in thickness after exposure to boiling water.

4.3.2 Resistance to light. Five tests shall be conducted from a 5-yard sample unit of the webbing. The test specimens shall be exposed in the accelerated weathering unit as specified in method 5804 of CCC-T-191. The specimens shall be placed side by side on the rack in such a manner that only the center portion of each specimen is exposed. Corex D filters and sunshine carbons shall be used. The exposure time shall be 50 hours. The spray heads shall be shut off during the entire exposure period. At the end of the exposure period, the specimens shall be brought to equilibrium under standard conditions as defined in CCC-T-191. They shall then be tested for breaking strength as specified in table IV and the percent of breaking strength (B.S.) loss shall be calculated as follows:

$$\frac{\text{Original B. S. - B. S. after aging}}{\text{Original B. S.}} \times 100 = \text{percent of B. S. loss.}$$

4.3.3 Resistance to heat. Five tests shall be conducted from a 5-yard sample unit of the webbing. The test specimens shall be suspended in a circulating air oven at a temperature of 180° ± 3°C. (356° ± 5°F.) for 1 hour. After removal from the oven, the specimens shall be brought to equilibrium under standard conditions as defined in CCC-T-191. They shall then be tested for breaking strength as specified in table IV and the percent of breaking strength loss shall be calculated as follows:

$$\frac{\text{Original B.S. - B.S. after aging}}{\text{Original B. S.}} \times 100 = \text{percent of B.S. loss.}$$

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.2).

5.1.1 Levels A, B and C. Tape, put up as specified, shall be packaged in accordance with the applicable requirements of MIL-W-43334.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. Tape shall be packed in accordance with the applicable requirements of MIL-W-43334.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-W-43334.

6. NOTES

6.1 Intended use. The tape is intended for use in the fabrication of canopies of ribbon parachutes.

6.2 Ordering data. Procurement documents should specify:

a. Title, number and date of this specification.
b. Length of roll (see 3.4).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

Custodians:

Navy - AS
Air Force - 11

Review activities:

Navy -
Air Force - 82, 85

Preparing activity:

Air Force - 11

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